Marco Del Fava

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GitHub | LinkedIn | Portfolio

Summary

Biomedical Engineering M.S. student at UC Davis working on human single-unit electrophysiology, neural acquisition, and Python analysis pipelines. Experience spans Intan/DeepArray bench testing, Neuralynx and Dark Horse Neuro workflows, spike sorting, experiment synchronization, and event-aligned response analysis. Develops tools with AI coding agents and evaluates outputs against recorded signals, timing measurements, and reproducible neural responses.

Research and Volunteer Experience

Research Volunteer — UC Davis Health — Z-Lab (Dr. Jie Zheng) | Sacramento, CA | Jul 2025 – Present

- Process human microwire recordings with SpikeInterface, MountainSort5, Kilosort4, and O-Sort; compare Neuralynx CSC and MED-derived inputs to investigate conversion precision and spike-detection discrepancies.
- Develop experiment synchronization with PsychoPy, Intan RHX TCP annotations, and Cedrus C-Pod TTL markers; measured clock drift and event-alignment residuals to assess timing limitations.
- Co-led bench testing of a 128-channel DeepArray/Intan acquisition setup; mapped probe and RHX channels and investigated grounding, filter behavior, shorts, and high-impedance channels. Lab testing identified 121 of 128 channels with clean test-signal recordings.
- Built a Python workflow combining sorted single-unit data and event markers to generate PSTHs for repeated 30-second stories; reproduced an event-associated response in the same unit across six repetitions.

Research Volunteer — UC Davis Health — Neuroscience Unit E5 | Sacramento, CA | Dec 2022 – Jul 2023

- Prepared headboxes and EEG application kits for the Epilepsy Monitoring Unit.
- Shadowed EEG technologists and observed clinical neurophysiology procedures for patients with epilepsy and traumatic brain injury.

Selected Projects

Semantic Neuron simulation | 2026

- Built a NEURON prototype linking sentence-embedding cosine similarity to stochastic synaptic input in a Hodgkin-Huxley compartment with fixed synaptic weights.
- Implemented seeded trials and outputs for membrane voltage, presynaptic and postsynaptic spike times, and firing rates to explore semantic-tuning hypotheses in simulation.
- Links: GitHub

Dark Horse Neuro / Neuralynx acquisition pipeline | 2026

- Develop Python orchestration for DHN-AQ configuration, SSH launch, acquisition diagnostics, and Neuralynx Pegasus/NetCom integration.
- Implemented RC/CS configuration writers and an NRD-to-UDP replay prototype; hardware reception and recording validation remain in progress.
- Links: GitHub · Docs

Experience

Operations Specialist — First Step Communities | Sacramento, CA | Jul 2021 – Present

- Analyzed program and partner datasets using Python, SPSS, and Excel to identify predictors of successful exits and service effectiveness.
- Coordinated data collection from clinical partners including UC Davis Health and Kaiser Permanente to support integrated care outcome analyses.

Analyst — Eurofins Test America | West Sacramento, CA | Oct 2019 – Jun 2021

- Maintained high-accuracy data entry and review in a regulated laboratory environment; supported QA documentation and adherence to standard operating procedures.
- Performed solvent extraction of organic compounds from soil and water samples in a high-throughput production setting.

Education

M.S. Biomedical Engineering | University of California, Davis | Sep 2026 - Present

Neuroengineering track; research in Dr. Jie Zheng's Z-Lab

M.S. Psychology (GPA: 3.7) | Arizona State University | 2023

B.S. Neuroscience, Physiology, and Behavior | University of California, Davis | 2017

Skills

Neural Data & Signal Processing: Human intracranial recordings and Behnke-Fried microwire data, Spike sorting (SpikeInterface, MountainSort5, Kilosort4, O-Sort), Preprocessing and waveform analysis, Event alignment and peri-stimulus time histograms (PSTHs)

Neural Acquisition & Tooling: Intan RHX and 128-channel DeepArray bench testing, Neuralynx CSC/NEV data and Pegasus/NetCom workflows, Dark Horse Neuro acquisition and MED conversion workflows, PsychoPy/TCP event integration and Cedrus C-Pod TTL markers, Channel mapping, grounding, and signal-quality troubleshooting

Scientific Software & Modeling: Python (NumPy, Pandas, Matplotlib, Neo), MATLAB and AWS GPU sorting workflows, NEURON and semantic embedding prototypes, Git, Linux, and shell scripting

Software Validation & Reproducibility: AI-assisted scientific software development, Recorded-data comparisons and repeatable response checks, Clock-drift and event-timing analysis, Synthetic signal and simulation-based prototyping

Certifications

- **Data Analysis with Matplotlib, Pandas, and Scikit-Learn Python Libraries** — UC Davis Continuing and Professional Education (2025-09-18) — View badge
- CITI — Biomedical Research
- CITI — Essentials of Research Administration
- CITI — GCP Social and Behavioral Research Best Practices